

Streaming Live RAG - Benchmark & Evaluation Report

Team: UNDEAD Theme: 4

This report distinguishes the local engineering benchmark from the official Samsung benchmark. Official acceptance-gate thresholds must be measured on the official Theme 4 test corpus when supplied.

1. Official Evaluation Gates

Gate	Target
G1 Reproducibility	Pass / Fail
G2 Early Retrieval	>= 80% of eligible queries
G3 Multi-Intent Identification	>= 70% of compound queries
G4 Factual Grounding	>= 85% citation support
G5 Session Refinement	Verified state continuity
G6 Telemetry & Observability	100% trace coverage

2. Local Engineering Smoke Test

Command: `python scripts/run_benchmark.py`

Case	Expected behavior	Observed local result
Early retrieval	Retrieve before final utterance	PASS - provisional retrieval
Multi-intent	At least two subqueries	PASS - 3 subqueries
Late refinement	No session restart	PASS - late_refinement trigger, answer version increments
Query suppression	No corpus retrieval for formatting request	PASS - NO_RETRIEVAL
Citation presence	Answer exposes retrieved sources	PASS - citations emitted

The local smoke test is intentionally small. It verifies that the architecture works end-to-end but does not establish performance on Samsung hidden tests.

3. Ablation Experiments

A. Hybrid vs sparse-only compares the configured dense+sparse score with sparse-only retrieval. The same query and corpus are used, and the top document ordering and scores are logged. This isolates the contribution of the dense component.

B. Controller vs always-retrieve demonstrates the controller's ability to suppress a presentation-only request after an answer already exists. The expected outcome is that no new retrieval call is made.

4. Metrics to Capture on the Official Test Set

Metric	Measurement method
Early retrieval rate	Eligible cases where RETRIEVE occurs before final transcript completion.
Multi-intent accuracy	Compound queries with at least the required distinct sub-intents correctly extracted.
Grounding rate	Sampled factual claims supported by a retrieved corpus citation.
Refinement continuity	Late-constraint cases preserving session state and incrementing answer version.
Retrieval latency	Milliseconds from controller trigger to fused evidence ready.
Retrieval suppression rate	Formatting-only turns correctly classified as NO_RETRIEVAL.
Telemetry coverage	Expected execution events present in trace logs.

5. Reproducibility Procedure

1) Build the Docker image. 2) Start API and UI with **docker compose up --build**. 3) Run unit tests with **pytest -q**. 4) Replace the demo corpus through CORPUS_PATH. 5) Replay the official benchmark cases. 6) Save telemetry and benchmark JSON artifacts. 7) Add measured figures to the final submission report.

6. Current Status

The repository is locally validated with automated tests and a small demonstration benchmark. The official Samsung corpus/test replay, final benchmark measurements, demo recording, and final Git tag require the team to perform those submission actions and are intentionally not fabricated.